

ASUS Zenbook S 16

AMD Ryzen AI 300 impresses with stellar performance across the board

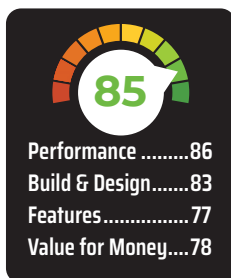


The Asus Zenbook S 16 is a special laptop

in many ways; it's one of the slimmest, lightest, and fastest 16-inch thin and light laptops around, and it's also running on the very latest AMD Zen 5 platform, which was announced at Computex 2024. We've covered the platform specifics via our deep-dive features on the all-new AMD XDNA 2 NPU, which is the star of the show as far as the new AMD Zen 5 architecture is concerned. We got a glimpse of the Asus Zenbook S 16 at the AMD Techday 2024 event in Las Vegas earlier this month. The Zenbook S 16 was one of our top picks of what we saw at the AMD Zen 5 Techday. Now that we've reviewed it, we can say with confidence that the Asus Zenbook S 16, with the shiny new AMD Ryzen AI 300 platform under its hood, is a delightful combination of performance, design, and efficiency.

CPU PERFORMANCE AND BENCHMARKS

The Asus Zenbook S 16 delivered an



impressive performance across the board, consistently producing some of the best scores we've seen from an x86-based thin and light laptop to date. However, it does exchange blows with the ARM-based Qualcomm Snapdragon X-Elite-based Asus Vivobook 15, particularly in the relatively new Cinebench 2024 benchmark, where the X-Elite surpasses its x86 rivals.

AI PERFORMANCE AND BENCHMARKS

We installed the Procyon Professional Benchmarking Suite and ran the Procyon AI Image Generation Benchmark, which utilizes the Stable Diffusion 1.5 (FP16) model. This benchmark generates 16 images in batches of four and returns a score based on the time taken to produce the images. Additionally, we tested AI performance by running the AI Computer Vision Benchmark, also part of the Procyon Suite.

As shown, the AMD Ryzen AI 9 HX370-based Asus Zenbook 16 demonstrates a distinct lead over the Sam-

sung Galaxy Book 4 Pro 16, powered by the Intel Core Ultra 7 155H processor. Despite having comparable power profiles with a base TDP of 28W, the AMD processor excels in graphics performance and benchmarks.

GRAPHICS PERFORMANCE AND BENCHMARKS

To evaluate graphics performance, we conducted a range of synthetic and real-world gaming benchmarks. The results below illustrate the AMD Zen 5 platform's significant strength in graphics performance, which translates to a notable advantage in both gaming and on-device generative AI workloads. Given that most on-device AI workloads and benchmarks focus on image generation capabilities, the AMD Ryzen AI 9 HX370-based Asus Zenbook 16 exhibits a clear lead in 720p and 1080p gaming across various genres. This performance advantage is also reflected in synthetic benchmarks like 3DMark. The Asus Zenbook 16 achieves an average of approximately 50FPS at 1080p with medium settings, showcasing the RDNA 3.5's impressive integrated GPU performance.

BATTERY LIFE

Typically, high performance comes at the cost of battery life, especially when it comes to good gaming performance. However, the Asus Zenbook S 16 defies this convention, offering great battery life alongside impressive iGPU performance. We tested the Zenbook S 16's battery life using the Procyon Video Playback battery life test, which drains the battery to 5% via a video playback loop test. The Zenbook S 16 delivered 11 hours and 33 minutes of battery life.

While not the best result we've seen, it's remarkably close to the top performer, the Asus Vivobook S 15, which runs on the Qualcomm Snapdragon X Elite platform. Unfortunately, we couldn't run the same benchmark on the Samsung Galaxy Book 4 Pro 16 due to persistent crashes, even after a fresh Windows installation. We've reported the issue to Procyon and will update the